



Chapter 2 – TCP/IP, UDP, TCP 3 Way Handshakes & OSI Model

1. Network Protocols

A **network protocol** is a set of rules that define how data is communicated between devices over a network.

The two main transport protocols are **TCP/IP** and **UDP**.

A. TCP/IP (Transmission Control Protocol / Internet Protocol)

What is TCP/IP?

- A **connection-oriented** protocol.
- Connection is established *before* sending data.
- Breaks data into **segments**.
- Performs **error checking** and **retransmission** if any data is lost.

How It Works?

1. Creates a reliable connection between sender and receiver.
2. Sends data in small segments.
3. Checks whether data reached correctly.
4. If any segment is lost, it is resent.

Why Is It Used?

- Because it is **reliable**, **accurate**, and **secure**.
- Ensures that data arrives correctly and in order.

Daily Life Example

- Uploading a video on YouTube.
(You can't afford missing video data, so reliability is important.)
-

B. UDP (User Datagram Protocol)

What is UDP?

- A **connectionless** protocol.
- No handshake, no guarantee.
- Sends data without checking errors.

How It Works?

- Sends packets called **datagrams** without confirming delivery.
- No retransmission even if data is lost.

Why Is It Used?

- Fast, low-latency communication is required.
- Some data loss is acceptable.

Daily Life Example

- Live streaming a cricket match.
- Online gaming.
- Video calls.

(If a few packets drop, your stream may blur for a second, but it continues without lag.)

2. TCP 3-Way Handshake

Before sending data, TCP creates a **secure and reliable connection** using a 3-step process.

Steps in 3-Way Handshake

Step 1 – SYN (Synchronize)

- Client sends a **SYN packet** to the server.
- Includes an initial sequence number, e.g., $x = 10$.

Step 2 – SYN + ACK (Synchronize + Acknowledge)

- Server receives SYN $\rightarrow$ increases the value by 1 $\rightarrow x + 1$.
- Server also sends its own sequence number, e.g., $y = 20$.
- Sends back: **SYN + ACK**.

Step 3 – ACK (Acknowledge)

- Client receives SYN+ACK $\rightarrow$ increases y by 1 $\rightarrow y + 1$.
- Sends ACK to server.

Now the connection is established.

Why Used?

- To create a reliable, synchronized connection between client and server.
- Ensures both sides are ready to send/receive data.

Daily Life Example

- When you open Instagram or Google $\rightarrow$ your device first establishes a TCP connection using this handshake.

3. OSI Model (7 Layers)

OSI = **Open Systems Interconnection** model.

It explains **how data travels from sender to receiver** in 7 steps.



Sender Side (Data going from User → Network → Cable)

1. Application Layer

What?

- The layer where humans interact with software.
- Browsers, email apps, WhatsApp, etc.

How it Works?

- User types a request like google.com.
- Application layer prepares the request but **cannot understand network language**.

Example:

- Opening a website in Chrome.
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2. Presentation Layer

What?

- Converts data into a standard format.
- Responsible for **encryption, compression, and translation**.

How it Works?

- Sender side: **Encrypts data**
- Receiver side: **Decrypts data**

Example:

- HTTPS encryption while logging into Gmail.
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3. Session Layer

What?

- Maintains the **communication session** between devices.

How it Works?

- Ensures that the session does not break while data transfer is happening.

Example:

- Your Zoom call staying connected for hours.
-

These top 3 layers = Software Layers

4. Transport Layer

What?

- Breaks big data into small pieces (segments).
- Adds **sequence numbers**.

How it Works?

- If you send a 10MB file, it is divided into many segments.
- Each segment gets a unique sequence number to reassemble later.

Example:

- Sending a file on WhatsApp.
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5. Network Layer

What?

- Works with **IP addresses**.
- Determines the best path for data.

How it Works?

- Router operates here.

- Adds sender and receiver **IP addresses**.

Example:

- Your phone sends data to YouTube servers using IP routing.
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6. Data Link Layer

What?

- Deals with **MAC addresses** and link-level communication.

How it Works?

- Converts IP packets → Frames.
- Adds MAC address for local delivery.

Example:

- Switch forwarding your laptop's data using MAC table.
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7. Physical Layer

What?

- Final layer where data is converted into **binary (0s and 1s)**.
- Travels through cable, Wi-Fi signals, fiber optics, etc.

Example:

- Electrical signals in LAN cable.
-
-



Receiver Side (Data coming from Cable → User)

The same layers work **in reverse** on the receiver side:

1. Physical Layer

- Converts electrical/wireless signals back to binary.

2. Data Link Layer

- Checks the **MAC address**.
- Ensures data is delivered to the correct device.

3. Network Layer

- Finds receiver's **IP address**.

4. Transport Layer

- Reassembles segments using **sequence numbers**.

5. Session Layer

- Maintains the communication session.

6. Presentation Layer

- Decrypts and translates data.

7. Application Layer

- Displays the final output to the user (webpage, video, message, etc.)



1 CHAPTER SUMMARY



Network Protocols

A **network protocol** is a **set of rules** that defines **how data is sent, received, and understood** between devices on a network.



Without protocols, devices cannot communicate properly.



Main Transport Layer Protocols:

- TCP (Transmission Control Protocol)
 - UDP (User Datagram Protocol)
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◆ A. TCP/IP (Transmission Control Protocol / Internet Protocol)



What is TCP?

- **Connection-oriented protocol**
 - Connection is established **before data transfer**
 - Data is broken into **segments**
 - Provides:
 - Error checking ✓
 - Retransmission ✓
 - Ordered delivery ✓
-



How TCP Works

- 1 Establishes a reliable connection
- 2 Breaks data into small segments
- 3 Sends segments with sequence numbers
- 4 Checks if data arrived correctly
- 5 Retransmits lost segments

📌 Guarantees **accuracy and reliability**

? Why TCP is Used?

- When **data loss is unacceptable**
- When **accuracy is more important than speed**

📱 Daily Life Example:

Uploading a video on **YouTube** 🎥

➡ Missing even small data can corrupt the video

◆ B. UDP (User Datagram Protocol)

🔍 What is UDP?

- **Connectionless protocol**
 - No handshake
 - No error checking
 - No retransmission
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⚙️ How UDP Works

- Sends data as **datagrams**
- No confirmation of delivery
- If data is lost → **ignored**

📌 Focuses on **speed**, not reliability

? Why UDP is Used?

- When **real-time speed** is important
- Some data loss is acceptable

🎮 Daily Life Examples:

- Live cricket streaming 🏏
- Online gaming 🎮
- Video calls 📺

➡ Small packet loss = blur, but no lag

🔄 TCP vs UDP (Quick Comparison)

Feature	TCP	UDP
Connection	Required	Not required
Reliability	High	Low
Speed	Slower	Faster
Error Checking	Yes	No
Use Case	File upload	Streaming, gaming

🔒 2 TCP 3-Way Handshake

Before sending data, TCP establishes a **secure and synchronized connection**.

📖 Steps of TCP 3-Way Handshake

● Step 1 – SYN

- Client sends **SYN** packet
- Includes initial sequence number ($x = 10$)

● Step 2 – SYN + ACK

- Server:
 - Acknowledges client $\rightarrow x + 1$
 - Sends its own sequence number ($y = 20$)

- Responds with **SYN + ACK**
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● Step 3 – ACK

- Client acknowledges server $\rightarrow y + 1$
 - Sends **ACK**
 -  Connection established
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? Why 3-Way Handshake is Used?

- ✓ Ensures both client & server are ready
- ✓ Synchronizes sequence numbers
- ✓ Prevents half-open connections

Example:

Opening **Google / Instagram** 

3 OSI Model (Open Systems Interconnection)

The **OSI Model** explains **how data travels** from sender to receiver using **7 layers**.

Sender Side (Top $\rightarrow$ Bottom)

1. Application Layer

- User interaction layer
- Browsers, email, WhatsApp

 Example: Typing google.com 

2. Presentation Layer

- Data formatting

- Encryption & compression

📌 Example: HTTPS encryption 🔒

🔗 3. Session Layer

- Manages communication session
- Keeps connection alive

📌 Example: Long Zoom meeting 🎥

👉 Top 3 = Software Layers 💻

📦 4. Transport Layer

- Breaks data into **segments**
- Adds sequence numbers

📌 Example: WhatsApp file sending 📁

🌐 5. Network Layer

- Works with **IP addresses**
- Finds best path

📌 Example: Router routing packets 🌐

🔄 6. Data Link Layer

- Uses **MAC addresses**
- Converts packets → frames

📌 Example: Switch forwarding frames 🔄

⚡ 7. Physical Layer

- Converts data into **0s & 1s**
- Transmits via cable/Wi-Fi

🔌 Example: LAN cable signals 🖱️

📡 Receiver Side (Bottom → Top)

Same layers work in **reverse order**:

- 1 Physical → signals to bits
 - 2 Data Link → MAC check
 - 3 Network → IP check
 - 4 Transport → reassembly
 - 5 Session → session control
 - 6 Presentation → decryption
 - 7 Application → display output
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🎯 4 CONCLUSION

This chapter builds the **core foundation of networking** 🔥

- **TCP & UDP** explain *how data moves*
- **3-Way Handshake** explains *how connections start*
- **OSI Model** explains *how data flows step-by-step*

👉 These concepts are **extremely important** for:

- ✓ Cyber Security
- ✓ Networking jobs
- ✓ Cloud & DevOps
- ✓ Technical interviews

Without understanding this chapter, **advanced networking & security is impossible** 🚫



5 MIND MAP

Lecture 2 – Networking

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|— Network Protocols

| |— TCP

| | |— Reliable

| | |— Connection-oriented

| | |— Error checking

| |

| |— UDP

| | |— Fast

| | |— Connectionless

| | |— No guarantee

|

|— TCP 3-Way Handshake

| |— SYN

| |— SYN + ACK

| |— ACK

|

|— OSI Model

|— Application

|— Presentation

|— Session

|— Transport

|— Network

└— Data Link

└— Physical

KNM



Q & A

? Q1. What is a network protocol?

✓ Answer:

A network protocol is a set of rules that defines how data is communicated between devices on a network.

? Q2. What is TCP?

✓ Answer:

TCP is a connection-oriented protocol that provides reliable, ordered, and error-checked data delivery.

? Q3. Why is TCP reliable?

✓ Answer:

Because it uses acknowledgements, sequence numbers, and retransmission.

? Q4. What is UDP?

✓ Answer:

UDP is a connectionless protocol that sends data without checking delivery or errors.

? Q5. TCP vs UDP?

✓ Answer:

TCP

UDP

Reliable

Fast

Connection-based

Connectionless

TCP

UDP

Error checking

No error checking

? **Q6. Where is UDP used?**

✓ **Answer:**

Live streaming, gaming, VoIP calls.

? **Q7. What is TCP 3-Way Handshake?**

✓ **Answer:**

It is a three-step process used by TCP to establish a secure and synchronized connection.

? **Q8. Steps of 3-Way Handshake?**

✓ **Answer:**

SYN → SYN+ACK → ACK

? **Q9. Why 3-Way Handshake is required?**

✓ **Answer:**

To ensure both client and server are ready and synchronized.

? **Q10. What is OSI Model?**

✓ **Answer:**

OSI Model is a 7-layer framework that explains how data travels from sender to receiver.

? **Q11. Which OSI layer does TCP/UDP belong to?**

✓ **Answer:**

Transport Layer.

? Q12. Which layer uses IP address?

✓ Answer:
Network Layer.

? Q13. Which layer uses MAC address?

✓ Answer:
Data Link Layer.

? Q14. Which layer handles encryption?

✓ Answer:
Presentation Layer.

? Q15. Which layer is closest to the user?

✓ Answer:
Application Layer.

? Q16. Which layer converts data to bits?

✓ Answer:
Physical Layer.

? Q17. Router works at which OSI layer?

✓ Answer:
Network Layer.

? Q18. Switch works at which OSI layer?

✓ Answer:

Data Link Layer.

? Q19. Why OSI model is important?

✓ Answer:

It helps in understanding, designing, and troubleshooting networks.

? Q20. Which protocol is used for file transfer?

✓ Answer:

TCP, because file transfer requires accuracy.
